

# Pulumi Assignment

## 1. S3 Bucket Creation

Create a Pulumi program to provision an S3 bucket with versioning enabled. Export the bucket name as output.

## 2. EC2 Instance with Security Group

Using Pulumi, create an EC2 instance along with a security group that allows:

- SSH (port 22)
- HTTP (port 80)

Export the public IP of the instance.

## 3. VPC and Networking Setup

Create a VPC with:

- One public subnet
- An Internet Gateway
- A route table

Ensure the subnet has internet access.

## 4. Multiple EC2 Instances

Write a Pulumi program to launch multiple EC2 instances dynamically (at least 2) using loops or configuration.

## 5. IAM Role Creation

Create an IAM role and attach a policy that allows read access to S3.

- Attach this role to the EC2 instance
- Restrict access to only the created S3 bucket

## 6. Outputs and Configuration

Use Pulumi outputs to display important resource details such as:

- Instance IDs
- Public IPs
- Bucket name

Ensure values are not hardcoded and are managed using configuration.

**Note**

1. Ensure all resources created using Pulumi have appropriate tags (e.g., Name, Environment).
2. Cloud provider credentials should be encrypted.